

A non-perturbative elastic network model of mutation

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Abstract

An elastic network model is specified by two sets of parameters, the spring constants k_{ij} and the equilibrium lengths l_{ij} . Standard practice fixes both from a single experimental structure, setting $l_{ij} = d_{ij}^e$ and $k_{ij} = f(d_{ij}^e)$, so that the potential is a function of that structure. Representing a mutation is then not straightforward. Perturbing the spring constants alone changes the dynamics but leaves the equilibrium structure unchanged, so a structural response requires perturbing the equilibrium lengths. This is what the linearly forced ENM (LFENM) does, treating the consequences perturbatively. Here we ask how far the same picture can be carried with no linearisation, no expansion, and no fixed contact map. We take the equilibrium lengths as the parameters, couple the spring constants to them through $k_{ij} = k(l_{ij})$, and define the mutant structure as the conformation that minimises the mutant potential. We derive two results for a single mutation from an arbitrary reference, which may itself be strained: a decomposition of the energy cost of a mutation into three terms, and the Hessian of a strained network, in which a transverse contribution absent from standard ENMs carries the effect of frustration on the dynamics. In both cases the extra terms are those that vanish for an unstrained reference with a fixed contact map.

1 Introduction

An elastic network model (ENM) represents a protein as beads joined by harmonic springs. Each spring is specified by two parameters, a spring constant k_{ij} and an equilibrium length l_{ij} , so the potential energy is

$$V(\mathbf{r}) = \frac{1}{2} \sum_{i < j} k_{ij} (d_{ij}(\mathbf{r}) - l_{ij})^2, \quad (1)$$

where $d_{ij}(\mathbf{r})$ is the distance between beads i and j in conformation $\mathbf{r}$. Despite its simplicity the model reproduces the slow collective motions of native proteins well.

Both parameter sets are usually read from one experimental structure $\mathbf{r}^e$: we set $l_{ij} = d_{ij}^e$ and take $k_{ij} = f(d_{ij}^e)$ for some decreasing function f , often a step at a cutoff distance. The potential is then a function of that structure. Two properties follow. The structure is an equilibrium conformation by construction, and the network is unstrained there, all springs sitting at their equilibrium lengths at once.

1.1 Representing a mutation

How should a mutation be represented in such a model? The two parameter sets play different roles. Perturbing the spring constants k_{ij} modifies the stiffness matrix and changes the normal modes, but leaves the equilibrium conformation where it was: a network whose springs already sit at their equilibrium lengths stays at equilibrium however the stiffnesses are scaled, so such a

perturbation cannot be used to predict conformational change [2]. For a mutation to **move the structure** it must perturb the **equilibrium lengths**.

This is the picture behind the linearly forced ENM (LFENM) [1, 2], in which a mutation perturbs the equilibrium lengths of the springs at the mutated site, $l_{ij} \rightarrow l_{ij} + \delta l_{ij}$, and the structural consequence is obtained perturbatively. Expanding the resulting change in energy **to first order in the displacements** recasts the perturbation as a set of forces directed along the perturbed contacts, $\mathbf{f}_{ij} = k_{ij} \delta l_{ij} \hat{\mathbf{e}}_{ij}$, with a reaction force at the mutated site that makes the total vanish. The mutant potential becomes harmonic-plus-linear, and the structural change follows in closed form as a linear response, $\delta \mathbf{r} = \mathbf{K}^{-1} \mathbf{f}$, with **$\mathbf{K}^{-1}$ the pseudo-inverse** taken over the **$3N - 6$ nonzero modes**. The **stiffness matrix** itself is left unchanged, so the mutant's normal modes are those of the reference. A self-consistent variant recovers them by taking the mutant structure predicted this way and rebuilding the network from it, which is how the effect of mutations on normal modes has been studied [3].

Two approximations enter here. The ENM potential is expanded to second order in the displacements, which discards a transverse contribution to the stiffness matrix and is exact only when every spring sits at its equilibrium length. **The energy change due to the mutation is then expanded to first order**, which is what produces the forces. **Both are controlled at an unstrained wild type. Neither is obviously controlled once mutations have accumulated.**

Here we ask how far **the same picture** can be carried with no linearisation. Every step below is exact: no expansion of the potential, no linear response, no truncation, and **no fixed contact map**. This is not to say the perturbative treatment is wrong. **Its accuracy and range of validity cannot be assessed from within itself**, and an exact formulation is what one compares against.

1.2 Two assumptions of the standard setup

Working exactly makes visible two assumptions built into the standard construction.

The first concerns the contact map. **Since k_{ij} is a function of the equilibrium distance d_{ij}^e** , which pairs are connected is a property of the structure. A mutation that changes equilibrium lengths then has no defined effect on the set of springs until the new structure is known, and the new structure cannot be computed without knowing the set of springs. One way out is to freeze the contact map, which forbids contacts from forming or breaking. Another is to iterate, predicting a structure with the springs one has and then rebuilding the springs from that structure. We avoid the **circularity** instead, by coupling the spring constant to the equilibrium *length* rather than to the distance,

$$k_{ij} = k(l_{ij}), \quad (2)$$

which makes the set of **active springs** a property of the parameters alone. A contact then forms or breaks when a mutation moves an equilibrium length across the range where k **falls off**, and this happens in the parameters, before any structure is computed. The parameters determine the structure and not the reverse.

The second concerns strain. Once a mutation has been applied, there is no reason the perturbed equilibrium lengths should be simultaneously realisable by any arrangement of beads in space, so the network settles at a compromise in which some springs remain stretched or compressed. Such a network is frustrated. Proteins are known to be frustrated, and frustration is thought to matter. Expressions derived for an unstrained network often contain terms that vanish there and are, for that reason, easy to omit without comment. Applied to a strained network they are no longer exact, and the discarded terms grow with the accumulated strain.

We take the wild type to be unstrained, since that is what the standard fit $l_{ij} = d_{ij}^e$ gives, but nothing below requires it. The theory is written for an arbitrary reference network, strained or not.

This matters for evolutionary trajectories, where each fixed mutant becomes the reference for the next mutation and is generally strained.

Open question. We assume an unstrained wild type only because the standard fit produces one. If frustration does affect the dynamics appreciably, a frustrated network might represent a native state better than an unstrained one, and we would want ways of fitting frustrated networks to a given structure. The Dresden notes list several candidate constructions.

1.3 Scope

We treat a single mutation from an arbitrary reference network. Section 2 defines the potential, the parameters and the state, and makes strain and frustration precise. Section 3 defines a mutation and the mutant structure. Section 4 derives the energy cost of a mutation and decomposes it exactly into three contributions. Section 5 derives the Hessian of a strained network, which governs its motions, and identifies the contribution through which frustration acts. Section 6 examines what changes when $k(l)$ is taken to be a step function. The substitution process itself, and the statistics of strain along an evolutionary trajectory, are not treated here.

Notation

N	number of beads, one per residue
$\mathbf{r}$	a conformation, the $3N$ coordinates $(\mathbf{r}_1, \dots, \mathbf{r}_N)$
$d_{ij}(\mathbf{r})$	$= \ \mathbf{r}_i - \mathbf{r}_j\ $, the distance between beads i and j
l_{ij}	equilibrium length of the spring on pair ij
Λ	the full set of equilibrium lengths $\{l_{ij}\}$
$k(l)$	spring constant as a function of equilibrium length, $k \geq 0$
K	the full set of spring constants $\{k(l_{ij})\}$, determined by Λ
$V(\mathbf{r}; \Lambda)$	the potential energy, eq. (3)
$\mathbf{r}^e$	an equilibrium conformation, where V attains its minimum
s_{ij}	$= l_{ij} - d_{ij}(\mathbf{r}^e)$, how far a spring sits from equilibrium
$\hat{\mathbf{e}}_{ij}$	$= (\mathbf{r}_i - \mathbf{r}_j)/d_{ij}$, the unit vector along pair ij
$\mathbf{I}$	the 3×3 identity

The subscripts *ref* and *mut* distinguish the reference network from the mutant. A subscripted $\mathbf{r}_{\text{ref}}^e$ or $\mathbf{r}_{\text{mut}}^e$ denotes the equilibrium conformation of that network: the conformation at which that network's own potential attains its minimum, which is in general a different conformation for each.

2 The model

2.1 The potential

The potential energy of a conformation $\mathbf{r}$ for a network with equilibrium lengths Λ is

$$V(\mathbf{r}; \Lambda) = \frac{1}{2} \sum_{i < j} k(l_{ij}) (d_{ij}(\mathbf{r}) - l_{ij})^2. \quad (3)$$

This is eq. (1) with the spring constants tied to the equilibrium lengths through eq. (2), so that Λ alone specifies the network. We take the zero of energy at the fully relaxed configuration, so V evaluated at any conformation is the strain energy stored in the springs.

The sum runs over all $\frac{1}{2}N(N-1)$ pairs. Pairs with $k(l_{ij}) = 0$ contribute nothing and are inactive; those with $k(l_{ij}) > 0$ are active, and the active pairs are the contact map. Since k is evaluated at l_{ij} rather than at $d_{ij}(\mathbf{r})$, membership of the contact map depends only on the parameters, whatever conformation V is evaluated at. The results below follow from this.

The function $k(l)$ is arbitrary and non-negative. We would normally take it decreasing, so that long springs are weak or absent, and if it falls to zero beyond some distance then that distance is a cutoff. A cutoff is a feature of a particular choice of k , not of the framework, and nothing below requires one.

2.2 States

A state of the model is an assignment of equilibrium lengths to every pair of sites, not only to the active ones. The wild-type state is obtained from an experimental structure $\mathbf{r}_{\text{wt}}^e$ in the standard way,

$$l_{ij}^{\text{wt}} = d_{ij}(\mathbf{r}_{\text{wt}}^e), \quad (4)$$

which reproduces the usual model: every spring sits at its equilibrium length at $\mathbf{r}_{\text{wt}}^e$, so that structure is a minimum of V with zero strain energy, and $k(l_{ij}^{\text{wt}}) = k(d_{ij}^e)$ recovers the usual assignment of spring constants.

Carrying equilibrium lengths for inactive pairs is required if contacts are to form. By eq. (4) every active wild-type pair has an equilibrium length inside the range where k is nonzero. If inactive pairs carried no parameter, none could ever be brought into that range, and mutation could only remove contacts. Giving every pair an equilibrium length makes formation and breaking symmetric. This costs nothing dynamically, since inactive pairs have $k = 0$ and contribute neither energy nor curvature: the Hessian keeps its $3N \times 3N$ form and the active pairs alone determine the motions.

2.3 Strain and frustration

Let $\mathbf{r}^e$ be an equilibrium conformation, where V attains its minimum, and define

$$s_{ij} \equiv l_{ij} - d_{ij}(\mathbf{r}^e), \quad (5)$$

the amount by which the spring on pair ij is compressed ($s_{ij} > 0$) or stretched ($s_{ij} < 0$) at equilibrium. The strain energy of the network is the value of the potential there,

$$E_{\text{strain}} = V(\mathbf{r}^e; \Lambda) = \frac{1}{2} \sum_{i < j} k(l_{ij}) s_{ij}^2. \quad (6)$$

A network is **frustrated** when the equilibrium conformation, which relaxes the network as far as it can be relaxed, still leaves some springs away from their equilibrium lengths. It follows that $E_{\text{strain}} > 0$.

The wild-type state of eq. (4) is unfrustrated: all s_{ij} vanish at $\mathbf{r}_{\text{wt}}^e$. A mutant is frustrated in general, since perturbed equilibrium lengths need not be simultaneously realisable by any arrangement of N points in three dimensions. Frustration is the generic case here and the unstrained network the exception, the reverse of the situation the standard construction is built for.

3 Mutation

A mutation at site m perturbs the equilibrium lengths of the springs attached to that site,

$$l_{ij}^{\text{mut}} = l_{ij}^{\text{ref}} + \delta l_{ij}, \quad \delta l_{ij} \sim \mathcal{N}(0, \sigma^2) \text{ for } i = m \text{ or } j = m, \quad (7)$$

leaving all others unchanged; σ sets the size of the mutation. This is the LFENM's representation of a mutation, and it is adopted here unchanged. What differs is what follows from it.

The spring constants are not independent parameters: they follow the equilibrium lengths through eq. (2), so the mutation induces

$$\delta k_{ij} = k(l_{ij}^{\text{ref}} + \delta l_{ij}) - k(l_{ij}^{\text{ref}}), \quad (8)$$

nonzero for any pair whose equilibrium length has moved into a region of different stiffness. Contacts form and break here, in the parameters, and nowhere else.

The mutant structure is the conformation at which the mutant potential attains its minimum,

$$\mathbf{r}_{\text{mut}}^e : V(\mathbf{r}_{\text{mut}}^e; \Lambda^{\text{mut}}) = \min_{\mathbf{r}} V(\mathbf{r}; \Lambda^{\text{mut}}), \quad (9)$$

and the mutant's energy is V evaluated there. Two features of this definition matter.

First, no expansion conditions the structure. Eq. (9) refers to the exact potential, not to a quadratic approximation of it. The mutant structure is not the reference displaced by a linear response to an equivalent force, and it is not obtained by truncating anything. Normal modes, when we compute them, describe a structure that has already been determined; they do not determine it.

Second, the contact map may differ between reference and mutant, by eq. (8). The comparison in Section 4 is then between two networks that need not have the same springs, and the bookkeeping has to accommodate springs present on one side and absent on the other.

4 The energy cost of a mutation

4.1 Three differences

Comparing a mutant with its reference involves two potentials and two conformations, so an energy difference is ambiguous until both are named. Three combinations are useful. The first is the cost of changing the network while holding the conformation fixed at the reference equilibrium,

$$\Delta V_{\text{stress}} = V(\mathbf{r}_{\text{ref}}^e; \Lambda^{\text{mut}}) - V(\mathbf{r}_{\text{ref}}^e; \Lambda^{\text{ref}}). \quad (10)$$

The second is the energy released when the mutant network relaxes to its own equilibrium,

$$\Delta V_{\text{relax}} = V(\mathbf{r}_{\text{mut}}^e; \Lambda^{\text{mut}}) - V(\mathbf{r}_{\text{ref}}^e; \Lambda^{\text{mut}}) \leq 0, \quad (11)$$

non-positive because $\mathbf{r}_{\text{mut}}^e$ is where $V(\cdot; \Lambda^{\text{mut}})$ attains its minimum. Their sum compares each network at its own equilibrium,

$$\Delta V_{\text{min}} = V(\mathbf{r}_{\text{mut}}^e; \Lambda^{\text{mut}}) - V(\mathbf{r}_{\text{ref}}^e; \Lambda^{\text{ref}}) = \Delta V_{\text{stress}} + \Delta V_{\text{relax}}. \quad (12)$$

The first compares two networks at one conformation, the second one network at two conformations, and the third is the difference of minima.

4.2 An exact decomposition

The first of these can be written out in full. Substituting $l_{ij}^{\text{mut}} = l_{ij}^{\text{ref}} + \delta l_{ij}$ and $k_{ij}^{\text{mut}} = k_{ij}^{\text{ref}} + \delta k_{ij}$ into eq. (10), and writing s_{ij} for the reference network's own departures from equilibrium, eq. (5), gives

$$\Delta V_{\text{stress}} = \underbrace{\frac{1}{2} \sum_{i < j} k_{ij}^{\text{ref}} \delta l_{ij}^2}_{(a)} + \underbrace{\sum_{i < j} k_{ij}^{\text{ref}} s_{ij} \delta l_{ij}}_{(b)} + \underbrace{\frac{1}{2} \sum_{i < j} \delta k_{ij} (s_{ij} + \delta l_{ij})^2}_{(c)}. \quad (13)$$

This is an identity. Nothing has been truncated, and it holds for any $k(l)$, any mutation size, and any reference network however strained. The three terms behave differently.

Term (a) is the work of stretching the springs that already exist. It is quadratic in the mutation size, always positive, and independent of the reference's strain. It is the only term that survives for an unstrained reference with an unchanged contact map.

Term (b) couples the mutation to the strain the reference already carries. It vanishes if and only if the reference is unstrained, all $s_{ij} = 0$, which is the case at the wild type by eq. (4). It is the only term of the three that can be negative: a mutation that happens to move equilibrium lengths toward the distances the structure already exhibits releases energy. Its size is proportional to the reference's strain, so it is negligible near the wild type and grows as the reference becomes more frustrated.

Term (c) is present only because the spring constants follow the equilibrium lengths. It vanishes if and only if every $\delta k_{ij} = 0$, which is what a fixed contact map assumes. Here a pair whose stiffness changes contributes its whole stored energy. Note that it is governed by $s_{ij} + \delta l_{ij}$, the departure the pair would have under its new equilibrium length, so a contact forming far from that length is expensive.

Terms (b) and (c) are what is discarded by a treatment set up at an unstrained reference with a frozen contact map. Neither is a higher-order correction to be recovered by expanding further. Both are contributions that vanish in that special case.

Open question. How large are (b) and (c) for a real protein, relative to (a) and to each other? The decomposition says when they vanish, not how much they matter. This is what the numerical work is for.

4.3 Differences, not energies

ΔV_{stress} is a difference, and is not interchangeable with the absolute energy $V(\mathbf{r}_{\text{ref}}^e; \Lambda^{\text{mut}})$ of the mutant network at the reference conformation. The two differ by $V(\mathbf{r}_{\text{ref}}^e; \Lambda^{\text{ref}})$, the reference's own strain energy, which is zero at the wild type and grows as the reference becomes frustrated. At the wild type the distinction is invisible. Away from it, it is the whole of the accumulated strain.

5 The dynamics of a strained network

5.1 The Hessian

Small motions about equilibrium are governed by the second derivatives of V , the Hessian, whose off-diagonal 3×3 blocks couple pairs of sites. Differentiating eq. (3) twice with respect to the coordinates gives, for an active pair,

$$\mathbf{K}_{ij} = -k(l_{ij}) \left[\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T + g_{ij} (\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T - \mathbf{I}) \right], \quad g_{ij} = \frac{l_{ij}}{d_{ij}(\mathbf{r})} - 1, \quad (14)$$

with the diagonal blocks fixed by translational invariance, $\mathbf{K}_{ii} = -\sum_{j \neq i} \mathbf{K}_{ij}$.¹ Inactive pairs drop out, so the matrix is $3N \times 3N$ however many pairs carry parameters. It has six zero eigenvalues, from the three translations and three rotations that leave V unchanged.

¹The sign of the transverse coefficient is easy to get wrong. The form above, with $g_{ij} = l_{ij}/d_{ij} - 1$, was checked against numerical second differences of eq. (3) at a frustrated minimum, and agrees to eight significant figures; the opposite sign does not.

5.2 The transverse contribution

Eq. (14) has two parts. The projector $\hat{\mathbf{e}}\hat{\mathbf{e}}^\top$ resists stretching along the line joining the two sites, and is the whole of the standard ENM Hessian. The part weighted by g_{ij} acts in the plane perpendicular to that line. Its coefficient measures the fractional mismatch between the spring's equilibrium length and its actual extension, and in terms of the departure from equilibrium is simply

$$g_{ij} = \frac{s_{ij}}{d_{ij}(\mathbf{r})}. \quad (15)$$

Grouping eq. (14) by projector makes the two directions explicit. Writing $\mathbf{P}_\perp = \mathbf{I} - \hat{\mathbf{e}}_{ij}\hat{\mathbf{e}}_{ij}^\top$ for the projector onto the plane perpendicular to the bond,

$$\mathbf{K}_{ij} = -k(l_{ij}) \left[\hat{\mathbf{e}}_{ij}\hat{\mathbf{e}}_{ij}^\top - g_{ij} \mathbf{P}_\perp \right], \quad (16)$$

so a pair has two effective force constants, not one: a longitudinal $k(l_{ij})$, independent of g_{ij} and always positive, and a transverse $-k(l_{ij})g_{ij}$, doubly degenerate, whose sign follows that of s_{ij} . For an isolated pair these are the two nonzero eigenvalues of the 6×6 Hessian, equal to $2k(l_{ij})$ and $-2k(l_{ij})g_{ij}$.

A stretched spring ($s_{ij} < 0$) resists transverse displacement, as a taut string does. A compressed one ($s_{ij} > 0$) assists it, and its transverse constant is negative: moving the two beads sideways lengthens the pair, which is what a compressed spring wants. The spring constant $k(l_{ij})$ itself is never negative. An isolated compressed pair is therefore unstable, while a compressed pair inside a network need not be, since the negative transverse curvature is one contribution among many.

The transverse part vanishes when every spring is at its equilibrium length. Frustration reaches the dynamics through g_{ij} and through nothing else: the departures s_{ij} enter the energy through eq. (6) and the motions through eq. (14), and nowhere else.

Standard ENMs omit the transverse part, and are entitled to, because their fit forces $l_{ij} = d_{ij}(\mathbf{r}^e)$ and hence $g_{ij} = 0$ at the conformation where the Hessian is evaluated. This is what the second-order expansion of the potential in the displacements accomplishes: keeping $(d_{ij} - l_{ij})^2 \approx (\hat{\mathbf{e}}_{ij} \cdot \mathbf{s}_{ij})^2$ retains the longitudinal projector and drops the rest. The omission is exact for an unstrained network and an approximation of uncontrolled sign for any other, including every mutant here.

We should separate this from term (b) of Section 4.2, since both vanish for an unstrained reference and both are proportional to the s_{ij} . They are different objects. Term (b) contributes to an energy difference and is linear in the mutation; g_{ij} contributes to the curvature of a single network and has nothing to do with any mutation. A network with no mutation applied to it can have $g_{ij} \neq 0$.

Open question. How far can s_{ij} grow before a compressed spring stops representing anything we mean by a residue contact? Nothing in the formalism breaks, and a network can carry compressed springs and remain stable, but a spring with l_{ij} well above d_{ij} is a strut under load rather than a weak contact. This bears on how large a mutation the model can be asked to describe.

6 The step-function case

Everything above holds for arbitrary $k(l)$. The common choice is a step at a cutoff $d_{\max}$,

$$k(l) = k_0 \mathbb{1}[l \leq d_{\max}], \quad (17)$$

so a pair is either a full-strength spring or absent. This makes V discontinuous in the parameters. An equilibrium length crossing $d_{\max}$ adds or removes a spring outright, and by term (c) of eq. (13) the energy jumps by $\frac{1}{2}k_0(s_{ij} + \delta l_{ij})^2$, which does not go to zero as the crossing becomes marginal. Two networks with nearly identical parameters can differ appreciably in energy.

Nothing in the preceding sections requires this. A smooth $k(l)$, a sigmoid of width w centred on $d_{\max}$ for instance, makes V continuous and differentiable in the parameters, keeps δk_{ij} small whenever δl_{ij} is small, and turns term (c) into a smooth contribution rather than a jump. The price is that the contact map stops being a binary object: every pair carries some stiffness, and the contact map becomes a thresholded description of a continuum. The two are ends of a modelling choice about how sharply stiffness falls with equilibrium length, and the theory accommodates either.

7 Discussion

The formulation departs from the standard ENM in one respect, coupling the spring constants to the equilibrium lengths rather than to the equilibrium distances. The contact map then becomes a property of the parameters, a mutation becomes a well-defined operation on the network that can create and destroy contacts, and the mutant structure becomes a derived quantity, obtained by minimising the mutant potential rather than by responding linearly to an equivalent force.

The two results, eq. (13) and eq. (14), show the same pattern. Each contains a contribution that is always present, plus contributions that vanish when the reference is unstrained with a fixed contact map. In the energy these are term (b), proportional to the accumulated strain, and term (c), arising from changes in stiffness. In the dynamics it is the transverse contribution g_{ij} . A treatment set up at an unstrained wild type sets all three to zero and is right to do so at that point. Our interest is in networks reached by mutation, where none of the three is zero and the discarded terms grow with the quantity that accumulates.

Both results are formal. They establish what the terms are and when they vanish, not how large they are for any particular protein.

What is not treated here: the substitution process, the statistics of strain along an evolutionary trajectory, and the selection models that would act on $\Delta V_{\min}$. Those are the layers this formulation is meant to support.

References

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